

Maitreyee Sarkar

📍 Indian Institute of Technology, Jodhpur

☎ +91 94 744 28 455

✉ sarkar.5@iitj.ac.in

🇮🇳 Indian

Experience

March 2021 – July Ongoing

Pursuing Ph.D. on Quantum Information and Computation

Indian Institute of Technology, Jodhpur (IIT J), Rajasthan, working with Prof. Atul Kumar, Department of Chemistry, IIT Jodhpur. Major aspect of my thesis is

- Application of quantum computing in quantum chemistry.

April 2023 – December 2023

Visiting Ph.D. program

International Institute of Information Technology, Hyderabad (IIIT H), Telangana.

- Computational molecular sciences with Machine Learning (ML).
- Generation of potential energy surfaces of molecules using ML techniques.
- Optimization of quantum circuits for quantum chemistry problems using ML techniques.

July 2019 – July 2020

M.Sc. Thesis on Quantum Information and Computation

Indian Institute of Technology, Jodhpur (IIT J), Rajasthan, was working with Dr. Atul Kumar, Associate Professor, Department of Chemistry, IIT Jodhpur. Major aspects of my thesis were

- To classify and quantify classical and quantum correlations in bi-partite and tri-partite states.
- Characterization of multiqubit entanglement and nonlocality by designing new Bell-type inequalities.

May 2016 – July 2016

Summer Research Fellowship Program (SRFP)

National Chemical Laboratory, Pune, under the supervision of Dr. Manjusha V. Shelke. The aspect of the internship was

- Incorporation of buffers (Zirconium Dioxide and Ferric Oxide) in CdS Quantum Dot Sensitized Solar Cells. This internship program was jointly organized by IAS (Bengaluru), INSA (New Delhi), NASI (Allahabad).

Publication

Aug 2026

Sarkar, M., Kumar, A. sym-ADAPT-VQE.

Communicated with the Journal.

July 2026

Sarkar, M., Raghavendra, V., Kumar, A. Assessing Optimal Optimizer for Quantum Simulation in Quantum Computers

<https://doi.org/10.48550/arXiv.2503.12084>

July 2026

Sarkar, M., Kumar, A. et al. Reaction Energies of Pericyclic Reactions using a Symmetry-Adapted VQE Algorithm

Communicated with the Journal.

June 2026

Sarkar, M., Roy, L., Gutal, A. et al. Symmetry-Match Fraction for Active-Space Selection: A Simple Route to Accurate Reaction Energies in VQE/UCCSD

<https://pubs.acs.org/doi/10.1021/acs.jpca.6c00363>.

June 2026

Roy, L., Sarkar, M., Tewari, M. et al. Reaction-Level Consistency within the Variational Quantum Eigensolver: Homodesmotic Ring Strain Energies of Cyclic Hydrocarbons

<https://doi.org/10.48550/arXiv.2603.12737>

April 2022	Singh, P., Faujdar, J., Sarkar, M. et al. Bell's inequality with biased experimental settings https://doi.org/10.1007/s11128-022-03504-x
Achievements	
May 31, 2023	IBM Quantum Challenge: Spring 2023 Achievement - Advanced https://www.credly.com/badges/e74371d3-e2e6-4f24-843a-8cb9c890639b/public_url
November 28, 2022	IBM Quantum Challenge Fall 2022 Achievement - Advanced https://www.credly.com/badges/023d0796-2f68-46a2-b7bc-3e00e88f220c/public_url
August 16, 2022	Qiskit Global Summer School 2022 - Quantum Excellence https://www.credly.com/badges/b91bd668-a8b0-4a44-8a4f-6aa5edeb2a2f/public_url
2016	Summer Research Fellowship Awardee Summer Research Fellowship jointly sponsored by Indian Academy of Sciences (IAS) Bengaluru, Indian National Science Academy (INSA) New Delhi, National Academy of Sciences, India (NASI) Allahabad.
2010	National Talent Search Examination (NTSE) Awardee NTSE Awardee organized by The National Council of Educational Research and Training (NCERT).
Education	
2021 – Ongoing	Ph.D. CGPA 9 IIT Jodhpur, Rajasthan, from Department of Chemistry
2018 – 2020	Master of Science, CGPA 8.8 IIT Jodhpur, Rajasthan, from Department of Chemistry
2014 – 2018	Bachelor of Science, CGPA 7.48 Presidency University, Kolkata, from Department of Chemistry
2013 – 2014	Higher Secondary Education, Percentage 82.8 West-Bengal Council of Higher Secondary education
2011 – 2012	Secondary Education, Percentage 92.43 West-Bengal Board of Secondary education
Presentation at Conference	
27 January, 2026	Poster Presentation 29th Annual Quantum Information Processing Conference, Riga, Latvia
17 September, 2025	Poster Presentation Theory of Quantum Computation, Communication and Cryptography, IISc Bengaluru, India
8 Dec, 2023	Poster Presentation At Theoretical Chemistry Symposium, IIT Madras, India
26 Jan, 2023	Oral Presentation At Progress in Quantum Science and Technology Conference, IIT Madras, India
Technical Strengths	
Operating system	Linux, Windows
Languages	Python, Fortran
Libraries	Numpy, Pandas, Pytorch, SciPy, Tensor Flow, Matplotlib
Software & Tools for quantum computing	Qiskit, PennyLane, MATLAB, Wolfram Mathematica, Latex
Packages & Tools for computational chemistry	Nwchem, Psi4, Avogadro, Orca, Gabedit, Molden
Relevant Courses	

Core Courses

- Quantum Machine Learning.
- Machine Learning and Deep Learning.
- Introduction to Quantum Information.
- Quantum Chemistry.
- Chemical Binding.
- Neumerical methods for group theory.

Other Courses

- Quantum Reality.
- Linear Algebra.
- Probability Theory.

Conference/Seminar Attended

26 Jan – 30 Jan, 2026	29th Annual Quantum Information Processing Conference At Atta Centre, Riga, Latvia
15 Sept – 19 Sept, 2025	Theory of Quantum Computation, Communication and Cryptography At Indian Institute of Science, Bengaluru, karnataka, India
7 Dec – 10 Dec, 2023	Theoretical Chemistry Symposium At Indian Institute of Technology Madras, Tamil Nadu, India
15 Feb – 18 Feb, 2023	Young Quantum conference At Harish-Chandra Research Institute, Prayagraj, Uttar Pradesh, India
23 Jan – 27 Jan, 2023	Progress in Quantum Science and Technology Conference At Indian Institute of Technology Madras, Tamil Nadu, India
18 Jul – 29 Jul, 2022	Qiskit Global Summer School Online Summer School on Quantum Simulations.
04 Oct – 08 Oct, 2021	Quantum Computing Online Elementary Faculty development Program, at ABV-Indian Institute of Information technology Management (ABV-IIITM), Gwalior, organized by AICTE Training and Learning (ATAL) Academy.
23 Aug – 27 Aug, 2021	Quantum Information Processing Applications Online Elementary Faculty development Program (FDP) on "Quantum Information Processing Applications" at Central University of Punjab, organized by AICTE Training and Learning (ATAL) Academy.
8 Dec – 11 Dec, 2019	National Conference on Quantum Information and Computations (QIC) At IIT Jodhpur, Rajasthan, organized by Interdisciplinary Research Program (IDRP)-QIC.

Teaching Assistantship

Aug – Nov, 2025	Hands-on session on Qiskit at IIT Jodhpur for MSc and Mtech
Jan – Apr, 2025	Computational Chemistry Lab at IIT Jodhpur for MSc and Mtech
Aug – Nov, 2024	Chemistry at IIT Jodhpur for Btech
Jan – Apr, 2024	B.Tech Lab at IIT Jodhpur for BTech
Aug – Nov, 2023	Chemistry at IIT Jodhpur for Btech
Jan – Apr, 2023	Seminal features of Quantum Information at IIT Jodhpur for Executive Mtech
Jan – Apr, 2023	Chemistry at IIT Jodhpur for Btech

Aug – Nov,2022	Quantum Machine Learning at IIT Jodhpur For MSc and MTech
Aug – Nov,2022	Foundation of Quantum Computing at IIT Jodhpur for Btech
Aug – Nov,2022	Numerical methods and group theory at IIT Jodhpur fot MSc
Jan – Apr,2022	Colloids and suspensions at IIT Jodhpur for MSc
Jan – Apr,2022	NMR-Principles at IIT Jodhpur for MSc
Aug – Nov,2021	Foundation of Quantum Information at IIT Jodhpur for Btech

Position Of Responsibility

July 2022 – Ongoing	Assistant Head Assistant Head at the Students Wellbeing Committee.
July 2022 – July 2023	General Secretary of Girl's Hostel General Secretary of Girl's Hostel Blue 2, under Board of Hostel Affairs.
July 2021 – June 2022	Student Guide Student Guide for at the Students Wellbeing Committee. • Student Guides help the students to be familiar with the institute and be connected with the counsellor to overcome any kind of psychological stress or issue.
July, 2019 – June, 2020	Students Elected Representative(SER) SER for M.Sc. 2nd year batch (Chemistry, Physics and Mathematics) at Students Elected Representative Society (SERS), IIT Jodhpur Gymkhana.
Aug, 2018 – June, 2019	Students Elected Representative(SER) Students Elected Representative for M.Sc. 1st year batch (Chemistry, Physics and Mathematics) at Students Elected Representative Committee (SERC), IIT Jodhpur Gymkhana.

Language proficiency

Written & Oral Proficiency	English, Bengali
Oral Proficiency	Hindi

Extra-Curricular Activities

2006, 07, 08	Senior Diploma, Bisharad, Ratna in Drawing and Painting Senior Diploma, Bisharad, and Ratna in Drawing and Painting From Bharatiya Sangeet and Sanskrito Samsad, West Bengal, India.
2009	Fourth Prize in Chess Tournament Chess Tournament Organized by Jalpaiguri District Cheess Association, West Bengal, India.